

Jeremiah J. McElwee

Earth and Environmental Sciences
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Research Statement

I am a geodynamicist who uses numerical modeling to study fluid-rock interaction. I am interested in finding ways to bridge micro-scale processes and macro-scale fluid migration in active tectonic and hydrothermal settings. I believe that geodynamic modeling benefits from multidisciplinary perspectives, and I work to incorporate observations from metamorphic petrology, experimental rock physics, and geophysics. I see a strong benefit in international collaboration and have experience working with international groups. My current work focuses on reaction-driven fracturing during (de)hydration reactions and seeks to apply the results to fluid migration and serpentinization in the oceanic lithosphere and the mantle wedge corner.

Education

University of Minnesota, Twin Cities Fall 2021 – Present
PhD, Geodynamics
Advisor: Ikuko Wada

University of California, San Diego Fall 2016 – Spring 2020
B.S., Physics with a minor in Earth Science

Research Experience

Graduate Research Assistant Fall 2021 – Present
EES at University of Minnesota, under Professor Ikuko Wada

Undergraduate Research Assistant Summer 2019
IGPP at Scripps Institute of Oceanography, under Professor Dave Stegman

Undergraduate Research Assistant Summer 2018
SERF at University of California, San Diego, under Doctor Carl Melis

Teaching Experience

Earth and Environmental Sciences Web Accessibility Coordinator Fall 2025

Lab Instructor

ESCI 2201 (Solid Earth Course for Geology Majors)

Fall 2021 – Fall 2025

ESCI 1003 (Introductory Paleontology Course for Non-Majors)

Spring 2022

Private Algebra and Geometry Tutor

Fall 2020 – Summer 2021

Peer Reviewed Publications

McElwee, J. J., Wada, I., Yoshida, K., Shimizu, H., & Okamoto, A. (2026). Impact of differential stress on fracture due to volume increasing hydration. *EGU: Solid Earth*, 17(3), 537–554. <https://doi.org/10.5194/se-17-537-2026>

Preprints and Manuscripts in Review or in Prep

McElwee, J.J., Wada, I., Yoshida, K., Okamoto, A., Effects of Reaction Weakening and Reaction-Induced Stresses on Fracture, In Review *JGR: Solid Earth*
<https://doi.org/10.22541/essoar.15004644/v1> [Preprint]

McElwee, J.J., Wada, I., Yoshida, K., Okamoto, A., Reaction-induced and tectonic stresses guide mantle wedge serpentinization, In Revision, To be Submitted to *Geochemistry, Geophysics, Geosystems*

Presentations and Workshops

“Effects of reaction weakening on reaction-induced fracturing: insights from numerical simulations”

Geo-processes in Mineral Carbon Storage Workshop 2026

Poster

“Impact of Reaction-Driven Fracturing on Fluid Flow and Hydration in the Lithosphere”

UMN Earth Student Research Symposium 2026

Oral

H43E “Localization of reaction-induced fracture and hydration due to reaction softening: insights from numerical simulations”

AGU Fall 2025

Poster

T43E “Effect of Confining Pressure on Reaction-Induced Fracture in the Mantle Wedge Corner”

AGU Fall 2025

Poster

“Reaction-induced fracture during hydration of the lithosphere: insights from numerical simulations”

Georgia Tech Solid Earth Seminar Fall 2025

Invited Oral

“Effects of Reaction Softening and Reaction-Induced Stresses on Fracture in the Mantle Wedge Corner”

CIG: The Next Generation Workshop Summer 2025

Poster

“Effects of Reaction Softening and Reaction-Induced Stresses on Fracture at the Base of the Mantle Wedge Corner” <i>Fluids in Cascadia CRESCENT Topical Workshop Spring 2025</i>	Poster
T53C-3247 “Effects of Reaction Softening on Reaction-Induced Fracture during Serpentinization of the Forearc Mantle Wedge Corner” <i>AGU Fall 2024</i>	Poster
“Tectonic and reaction-induced stresses control fracture, fluid flow, and serpentinization in the mantle wedge corner” <i>Montreal Fluids Workshop Spring 2024</i>	Poster
T52B-04 “Serpentinization in the Forearc Mantle Wedge: Effects of Tectonic and Reaction-Induced Stresses on Fracture Propagation and Fluid Flow” <i>AGU Fall 2023</i>	Oral
OC2-12 “Mantle wedge serpentinization in subduction zones: Effects of reaction-induced and tectonic stresses on fracture propagation and fluid flow” <i>WRI17/AIG14 Summer 2023</i>	Oral
T32E-0226 “Propagation of Serpentinization in the Mantle Wedge Corner” <i>AGU Fall 2022</i>	Poster
Attendee <i>SEG-AGU Geophysics of Convergent Margins Summer 2022</i>	Attendee

Collaborations

Ikuko Wada (PhD Advisor), University of Minnesota – Twin Cities, USA

Atsushi Okamoto, Tohoku University, Japan

Kazuki Yoshida, High Energy Accelerator Research Organization (KEK), Japan

Awards and Fellowships

Earth and Environmental Sciences Outstanding Teaching Assistant	2024 – 2025
Earth and Environmental Sciences Harold Mooney Fellowship and Banerjee Fellowship	2024 - 2025
Earth and Environmental Sciences Richard Clarence Dennis Graduate Fellowship	2022 – 2023

Service

Earth and Environmental Sciences Confronting Colonization Working Group
Fall 2024 – Summer 2025

Earth and Environmental Sciences Incoming Graduate Student Mentor
Fall 2023 – Present

Earth and Environment Display Committee
Fall 2022 - Summer 2025

Graduate Student Representative to Earth and Environmental Sciences Faculty Meeting
Fall 2023 - Summer 2024

Earth and Environmental Sciences Outreach Committee
Fall 2021 - Summer 2023

Elementary School, Middle School, and Museum Science Education Activities
Intermittent

Earth Science Activities for Summer Camps
Summer 2022

References

Ikuko Wada: iwada@umn.edu

Lars Hansen: lnhansen@umn.edu

Atsushi Okamoto: atsushi.okamoto.d4@tohoku.ac.jp